

Chapter 11: Probability

5 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Statistics and probability: MA4-PRO-C-01 solves problems involving the probabilities of simple chance experiments.

CHAPTER

11

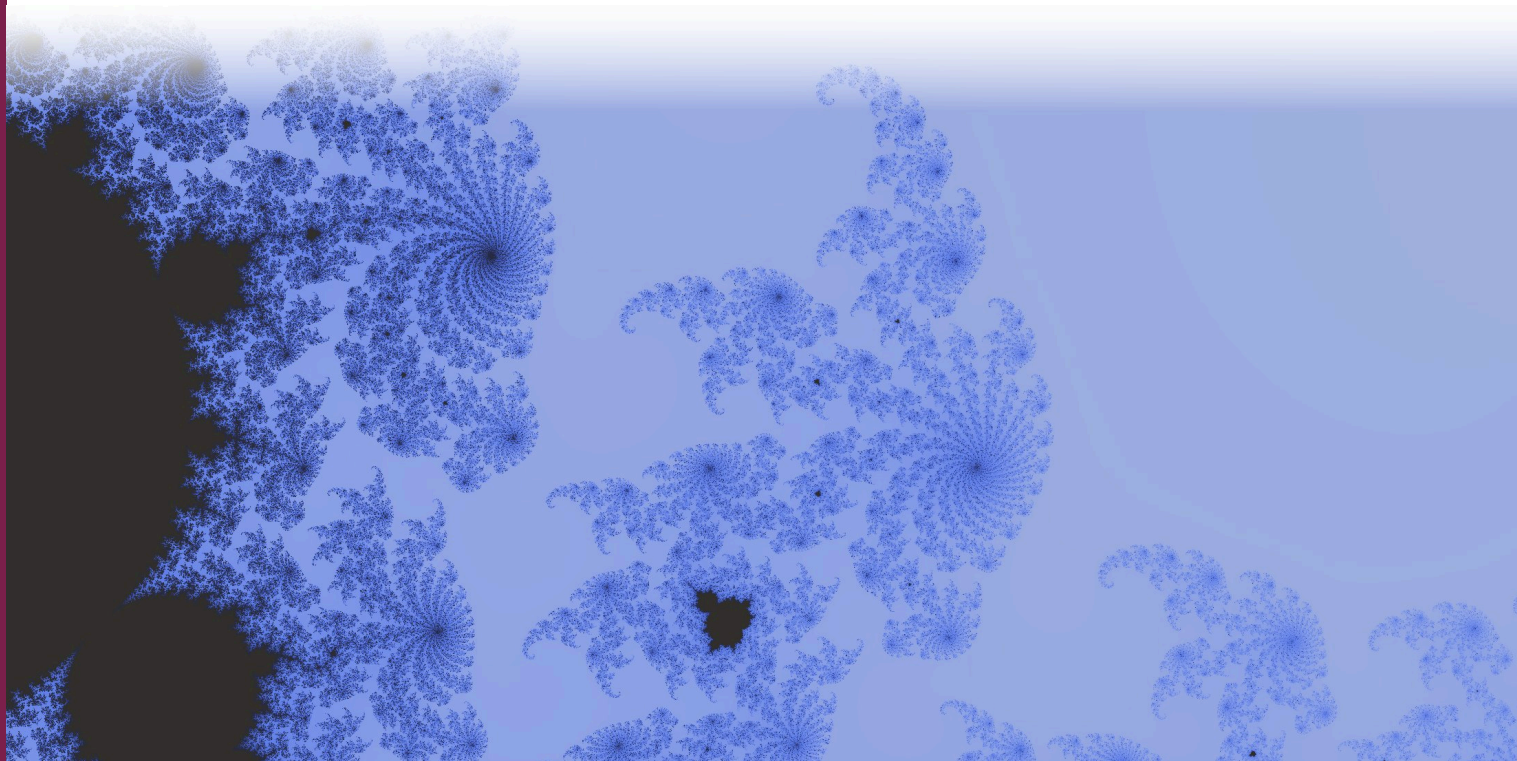
11.01 Sample space

11.02 Probability

11.03 The probability scale

11.04 Experimental probability

11.05 Complementary events



How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

WE DO **Grey boxes.** Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

YOU DO **Blue boxes.** Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

DRILL **Skill drill pages.** After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

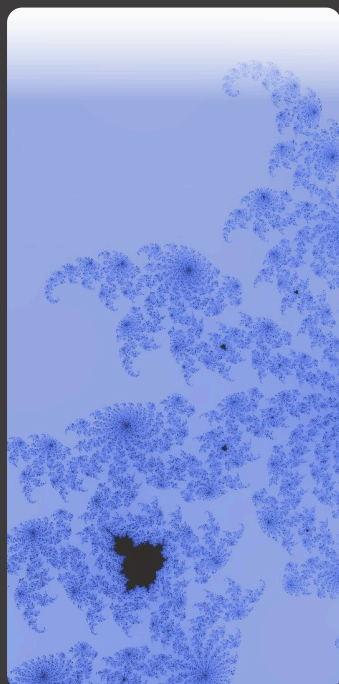
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
11.01	Sample space	3		☐
11.02	Probability	9		☐
11.03	The probability scale	15		☐
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In this chapter you will

- list the sample space of an experiment
- find the probability of simple events as fractions, decimals and percentages
- use the probability scale and chance words
- compare experimental and theoretical probability
- find the probability of complementary events



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is feathery spirals deep in Elephant Valley. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
11.01

Sample space

Date _____

Learning intentions

We are learning to:

- list the sample space of a chance experiment
- decide whether outcomes are equally likely

Success criteria

I can:

- ☐ list every possible outcome
- ☐ write the sample space in braces { }
- ☐ explain why outcomes are or are not equally likely

Key terms Write the definition as your teacher explains it.

Outcome

Sample space

Equally likely

Notes

Level 1 · Getting started

Easy · 17 questions

WE DO With your teacher: copy what they write on the board.

Example 1 List the sample space.

a List the sample space for tossing a coin.

Your teacher works this out. Copy every step.

b List the sample space for rolling a die.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: List the sample space.

12 QUESTIONS

REMEMBER An **OUTCOME** is one possible result.

1 Spinning a spinner with A, B, C, D

2 Choosing a day of the week

3 Choosing a vowel

4 Rolling a die

5 The result of a soccer match for one team

6 The season of a person's birthday

Questions 7–12: how many outcomes are in the sample space?

7 tossing a coin

8 rolling a die

9 choosing a month

10 choosing a letter of the alphabet

11 spinning A, B, C, D, E

12 choosing a card suit

SKILL DRILL 1 Easy basics: practise until it is automatic

25 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 25 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–15: how many outcomes are in the sample space?

1 choosing a vowel

2 choosing an even number from 1 to 10

3 choosing a card from a deck

4 choosing a letter of the alphabet

5 choosing a colour of the rainbow

6 rolling an 8-sided die

7 choosing a suit of cards

8 choosing a month

9 tossing a coin

10 choosing a digit (0 to 9)

11 choosing a season

12 choosing a day of the week

13 choosing a letter from MATHS

14 rolling a die

15 spinning a spinner with 5 equal sectors

ROUND 2 Questions 16–25: list the sample space.

16 choosing a letter from BOOK

17 a spinner with red, blue, green

18 rolling a die

19 choosing a prime less than 12

20 choosing an odd number less than 10

21 choosing a suit

22 tossing a coin

23 choosing a letter from CAT

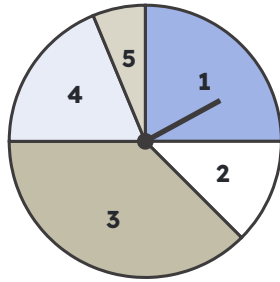
24 choosing a weekend day

25 choosing a vowel

YOU DO

On your own: **Easy** Set B. Use the spinner.

5 QUESTIONS



- 1 How many outcomes?
- 2 Why aren't they equally likely?
- 3 Which number is most likely?
- 4 Which is least likely?
- 5 Which number has a probability of 25%?



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

List all the possible outcomes.

- a** List the outcomes of a driving test.

Your teacher works this out. Copy every step.

- b** List the classifications of a new film.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. List the sample space. Are the outcomes equally likely?

6 QUESTIONS

REMEMBER The **SAMPLE SPACE** is the list of all outcomes.

- 1 The public transport students use to get to school

- 3 Choosing a letter from the word APPLE

- 5 The colour of a traffic light when you arrive

- 2 Choosing a letter from the word MATHS

- 4 The number of heads when 2 coins are tossed

- 6 Drawing a card suit from a deck

SKILL DRILL 2 Medium basics: keep practising

22 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 22 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–12: how many outcomes are there for both together? (Multiply.)

1 toss two coins

2 choose a month and a day of the week

3 roll a die and spin a 3-sector spinner

4 choose a day of the week and toss a coin

5 toss three coins

6 choose a suit and toss a coin

7 spin a 4-sector spinner and toss a coin

8 roll two 4-sided dice

9 choose an entrée (2), main (5) and dessert (3)

10 toss a coin and roll a die

11 choose a shirt (3) and pants (4)

12 roll two dice

ROUND 2 Questions 13–22: are the outcomes equally likely? Write yes or no.

13 winning or losing a footy match

14 spinning a spinner with equal sectors

15 spinning a spinner with one large sector

16 tossing a fair coin

17 whether it rains tomorrow

18 passing or failing a test

19 the gender of the next baby born

20 rolling a fair die

21 choosing a card from a shuffled deck

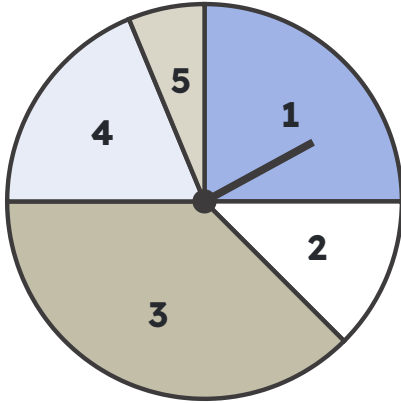
22 a bag with 5 red and 1 blue: the colour

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use the spinner, then list the outcomes.



a How many outcomes? Are they equally likely?

Your teacher works this out. Copy every step.

b List the sample space for the winner when Katrina plays Biljana at tennis.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Two-step experiments.

4 QUESTIONS

REMEMBER Write it in braces: {H, T}.

1 List the sample space for tossing 2 coins (e.g. HT).

3 How many outcomes are there for rolling 2 dice? Explain.

2 List all outcomes for spinning spinner ABCD and tossing a coin.

4 Design a spinner with 4 outcomes that are not equally likely.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

List the sample space for choosing a primary colour.

Level 2

How many outcomes are there when rolling a die?

Level 3

Are the outcomes of a test (pass/fail) equally likely? Explain.

 Extension · Stretch yourself Worked example

How many outcomes are there when 3 coins are tossed? List them.

1. Each coin has 2 outcomes: $2 \times 2 \times 2$
2. HHH, HHT, HTH, HTT, THH, THT, TTH, TTT

Answer: 8 outcomes

YOU DO

On your own: read the worked example, then try these.

- E1 How many outcomes when 4 coins are tossed?

- E2 How many different 2-letter codes can be made from A, B, C if letters can repeat?

Lesson summary

Missed the lesson? Start here.

Remember

1. An **outcome** is one possible result.
2. The **sample space** is the list of all outcomes.
3. Write it in braces: {H, T}.
4. Outcomes are **equally likely** if each has the same chance.

 Level 1

Tossing a coin

1. 2 outcomes

Answer: {H, T}

 Level 2

Rolling a die

1. 6 outcomes

Answer: {1, 2, 3, 4, 5, 6}

 Level 3

Spinner with unequal sectors

1. Different sector sizes

Answer: not equally likely

 Extension

3 coins

1. $2 \times 2 \times 2$

Answer: 8 outcomes

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LESSON
11.02

Probability

Date _____

Learning intentions

We are learning to:

- find the probability of a simple event

Success criteria

I can:

- ☐ use $P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total outcomes}}$
- ☐ write probability as a fraction, decimal or percentage
- ☐ round percentages to 1 decimal place

Key terms Write the definition as your teacher explains it.

Probability

Event

At random

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 A die is rolled. Find each probability.

a On a die, find $P(1)$.

b On a die, find $P(\text{odd number})$.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: A die is rolled. Find the probability of rolling: **12 QUESTIONS**

REMEMBER $P(\text{event}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$.

1 a 6

2 an even number

3 a number less than 5

4 a 7

5 a number greater than 0

6 a 2 or a 3

Questions 7–12: a bag has 3 red, 5 blue and 2 green counters. Find:

7 $P(\text{red})$

8 $P(\text{blue})$

9 $P(\text{green})$

10 $P(\text{red or green})$

11 $P(\text{yellow})$

12 $P(\text{not blue})$

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: find the probability as a fraction in simplest form.

1 7 red, 3 green: $P(\text{red})$

2 6 blue, 4 green: $P(\text{blue})$

3 8 red, 1 yellow: $P(\text{red})$

4 2 black, 1 white: $P(\text{black})$

5 4 red, 8 black: $P(\text{red})$

6 3 red, 9 yellow: $P(\text{red})$

7 2 red, 1 blue: $P(\text{red})$

8 1 blue, 4 green: $P(\text{blue})$

9 8 blue, 2 red: $P(\text{blue})$

10 3 yellow, 4 green: $P(\text{yellow})$

11 3 white, 6 black: $P(\text{white})$

12 1 red, 5 green: $P(\text{red})$

13 8 black, 9 green: $P(\text{black})$

14 4 black, 3 blue: $P(\text{black})$

15 6 yellow, 2 white: $P(\text{yellow})$

16 1 yellow, 4 black: $P(\text{yellow})$

17 2 yellow, 8 red: $P(\text{yellow})$

18 6 blue, 6 red: $P(\text{blue})$

19 6 red, 2 white: $P(\text{red})$

20 9 white, 4 red: $P(\text{white})$

21 4 black, 5 white: $P(\text{black})$

22 1 red, 2 green: $P(\text{red})$

23 9 green, 4 blue: $P(\text{green})$

24 7 green, 5 white: $P(\text{green})$

ROUND 2 Questions 25–40: a fair die is rolled. Find the probability.

25 $P(\text{a multiple of 3})$

26 $P(\text{less than 3})$

27 $P(\text{a 7})$

28 $P(\text{more than 4})$

29 $P(\text{an even number})$

30 $P(\text{a prime})$

31 $P(\text{a 4})$

32 $P(\text{less than 10})$

33 $P(\text{a square number})$

34 $P(\text{a factor of 6})$

35 $P(\text{a 1 or a 2})$

36 $P(\text{a number from 1 to 6})$

37 $P(\text{an odd number})$

38 $P(\text{not a 6})$

39 $P(\text{a multiple of 5})$

40 $P(\text{at least 5})$

YOU DO

On your own: **Easy** Set B. A letter is chosen at random from PROBABILITY. Find the probability of choosing:

6 QUESTIONS

1 B

4 Y

2 I

5 a letter in the word ROB

3 a vowel

6 Z



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Find each probability.

a On a die, find $P(\text{number less than 5})$.

Your teacher works this out. Copy every step.

b A box has 15 dark, 13 milk and 12 white chocolates. Find $P(\text{white})$.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Find each probability as a fraction and a percentage (1 d.p.).

6 QUESTIONS

REMEMBER Probabilities are between 0 and 1.

1 15 dark, 13 milk, 12 white chocolates: $P(\text{white})$

3 $P(\text{not milk})$

5 $P(\text{exercise books or paper})$

2 $P(\text{dark})$

4 Truck (1040 boxes): $P(\text{rulers})$

6 $P(\text{erasers})$

SKILL DRILL 2 Medium basics: keep practising

27 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 27 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–15: a letter is chosen at random from the word. Find the probability.

1 AUSTRALIA: P(U)

2 PARALLEL: P(A)

3 BANANA: P(A)

4 KINGSLIFF: P(K)

5 PARALLEL: P(P)

6 PROBABILITY: P(Y)

7 KINGSLIFF: P(N)

8 STATISTICS: P(A)

9 PARALLEL: P(L)

10 PROBABILITY: P(T)

11 KINGSLIFF: P(C)

12 STATISTICS: P(I)

13 AUSTRALIA: P(L)

14 AUSTRALIA: P(T)

15 MATHEMATICS: P(T)

ROUND 2 Questions 16–27: write the probability as a fraction, a decimal and a percentage.

16 4 chances in 5

17 8 chances in 25

18 1 chances in 2

19 19 chances in 20

20 9 chances in 10

21 3 chances in 25

22 1 chances in 4

23 7 chances in 10

24 3 chances in 4

25 23 chances in 25

26 3 chances in 20

27 9 chances in 25

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find each probability as a percentage (1 d.p.).

- a** A truck has 325 boxes of books, 210 of rulers, 360 of paper and 145 of pens. Find $P(\text{rulers})$ as a % (1 d.p.).

Your teacher works this out. Copy every step.

- b** Using the truck, find $P(\text{no pens})$ as a % (1 d.p.).

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Problems.

4 QUESTIONS

REMEMBER Simplify fractions.

- 1** A class has 14 girls and 11 boys. One student is chosen at random. Find $P(\text{boy})$ as a decimal.

- 3** A bag has red and blue counters. $P(\text{red}) = 0.35$. What is $P(\text{blue})$?

- 2** A raffle sells 250 tickets. You buy 4. Find your probability of winning as a percentage.

- 4** A bag has 12 counters and $P(\text{green}) = \frac{1}{4}$. How many are green?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find $P(5)$ on a die.

Level 2

A bag has 3 red and 7 blue marbles. Find $P(\text{red})$.

Level 3

Write your answer to Question 2 as a percentage.

 Extension · Stretch yourself Worked example

A card is drawn from a standard deck of 52.
Find $P(\text{red king})$.

1. There are 2 red kings (hearts, diamonds)
2. $\frac{2}{52}$ simplifies

Answer: $\frac{1}{26}$

YOU DO

On your own: read the worked example,
then try these.

- E1 Find $P(\text{a heart or a king})$ from a standard deck.

- E2 Find $P(\text{a picture card: J, Q, K})$ from a standard deck.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. $P(\text{event}) = \text{favourable outcomes} \div \text{total outcomes}$.
 3. Simplify fractions.

2. Probabilities are between 0 and 1.
4. Percentage: $\times 100\%$, round as asked.

 Level 1

$P(\text{odd})$ on a die

1. 3 odd of 6

Answer: $\frac{1}{2}$

 Level 2

$P(\text{white chocolate})$

1. 12 of 40

Answer: $\frac{3}{10}$ (C)

 Level 3

$P(\text{rulers})$

1. $210 \div 1040$

Answer: 20.2%

 Extension

$P(\text{red king})$

1. 2 of 52

Answer: $\frac{1}{26}$

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LESSON
11.03

The probability scale

Date _____

Learning intentions

We are learning to:

- describe chance using words and the probability scale

Success criteria

I can:

- ☐ place events on a scale from 0 (impossible) to 1 (certain)
- ☐ match probabilities to words such as unlikely and good chance
- ☐ find probabilities from a jar of objects

Key terms Write the definition as your teacher explains it.

Certain

Impossible

Even chance

Notes

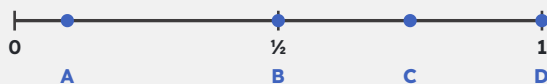
Level 1 · Getting started

Easy · 12 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the probability scale and chance words.

- a** Where on the scale is the chance that the next baby is a girl?



Your teacher works this out. Copy every step.

- b** Which word describes a probability of 0.29?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Describe each event using a chance word.

6 QUESTIONS

REMEMBER Probability goes from 0 (impossible) to 1 (certain).

- | | | |
|---|--|---|
| 1 The sun will rise tomorrow.
_____ | 2 Rolling a 7 on a die.
_____ | 3 A coin lands on heads.
_____ |
| 4 It will snow in Sydney in January.
_____ | 5 Rolling a number less than 6.
_____ | 6 Your teacher is over 10 years old.
_____ |

SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 36

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: write the chance word: impossible, unlikely, even chance, likely or certain.

1 $\frac{4}{5}$

2 $\frac{3}{5}$

3 25%

4 0

5 100%

6 40%

7 $\frac{19}{20}$

8 0.3

9 0.7

10 0.95

11 60%

12 90%

13 0.6

14 20%

15 $\frac{1}{2}$

16 0.5

17 $\frac{1}{4}$

18 $\frac{1}{20}$

19 0.9

20 $\frac{3}{4}$

21 10%

22 0.75

23 30%

24 1

ROUND 2 Questions 25–36: which chance word describes the event?

25 picking a vowel from ORANGE

26 winning the lottery

27 rolling a number less than 6

28 picking an ace from a deck

29 tossing heads

30 rolling a 7 on a normal die

31 a baby born on a weekday

32 choosing a red card from a deck

33 the sun rises tomorrow

34 rolling a number from 1 to 6

35 snow in Kingscliff in summer

36 rolling a 6

YOU DO

On your own: **Easy** Set B. Match each probability to a word: impossible, unlikely, even chance, likely, certain. **6 QUESTIONS**

1 0

4 0.85

2 0.29

5 1

3 $\frac{1}{2}$

6 15%



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Find each probability.

- a** Which is certain: choosing a 4 from {1, 2, 3, 4}; a red ball from green balls; an odd or even number on a die; a baby being a girl?

Your teacher works this out. Copy every step.

- b** A jar has 1 red, 6 yellow, 2 white and 5 black jellybeans. Find $P(\text{yellow})$.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. A jar has 1 red, 6 yellow, 2 white and 5 black jellybeans. Find:

6 QUESTIONS

REMEMBER $\frac{1}{2}$ is an even chance.

1 $P(\text{yellow})$

3 $P(\text{blue})$

5 $P(\text{black})$

2 $P(\text{white or red})$

4 $P(\text{not green})$

6 $P(\text{not yellow})$

SKILL DRILL 2 Medium basics: keep practising

28 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 28 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: which is more likely? Change to decimals to compare.

1 $\frac{5}{8}$ or 20%

2 $\frac{2}{5}$ or 35%

3 $\frac{2}{5}$ or 45%

4 $\frac{3}{8}$ or 10%

5 $\frac{1}{4}$ or 45%

6 $\frac{2}{5}$ or 10%

7 $\frac{5}{8}$ or 5%

8 $\frac{1}{3}$ or 40%

9 $\frac{3}{5}$ or 35%

10 $\frac{7}{10}$ or 15%

11 $\frac{1}{4}$ or 65%

12 $\frac{5}{8}$ or 80%

13 $\frac{5}{8}$ or 30%

14 $\frac{1}{4}$ or 20%

15 $\frac{1}{3}$ or 30%

16 $\frac{1}{4}$ or 90%

ROUND 2

Questions 17–28: a bag of counters is described. Is picking red impossible, unlikely, even chance, likely or certain?

17 5 red, 0 blue

18 2 red, 3 blue

19 1 red, 8 blue

20 7 red, 5 blue

21 2 red, 6 blue

22 5 red, 4 blue

23 1 red, 6 blue

24 1 red, 0 blue

25 2 red, 1 blue

26 6 red, 1 blue

27 2 red, 5 blue

28 4 red, 4 blue

Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find each probability.

a Find $P(\text{white or red})$.

Your teacher works this out. Copy every step.

b Find $P(\text{blue})$ and $P(\text{not green})$.

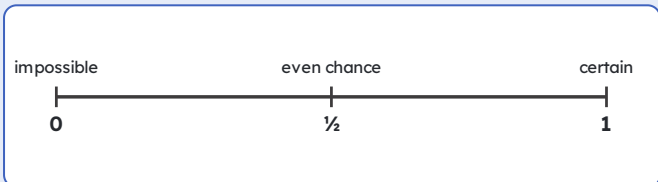
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Mark each event on the scale.

2 QUESTIONS

REMEMBER Less than $\frac{1}{2}$: unlikely. More than $\frac{1}{2}$: likely.

1 Mark: A rolling an even number; B rolling a 6; C rolling a number from 1 to 6; D rolling a 0.



2 Describe an event that is: (a) impossible (b) unlikely (c) certain.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What is the probability of an impossible event?

Level 2

Which word describes $P = 0.9$?

Level 3

A bag has 3 red and 5 blue. Find $P(\text{not red})$.

Extension · Stretch yourself

Worked example

A bag has 20 counters. $P(\text{red}) = 0.4$ and $P(\text{blue}) = 0.35$. The rest are green. How many are green?

1. $P(\text{green}) = 1 - 0.4 - 0.35 = 0.25$

2. 0.25×20

Answer: 5

YOU DO

On your own: read the worked example, then try these.

- E1** A spinner has $P(\text{red}) = \frac{1}{3}$. It has 12 equal sectors. How many are red?

- E2** Design a bag of 10 marbles where $P(\text{blue})$ is “unlikely” and $P(\text{red})$ is “likely”.

Lesson summary

Missed the lesson? Start here.

Remember

1. Probability goes from 0 (impossible) to 1 (certain).
3. Less than $\frac{1}{2}$: unlikely. More than $\frac{1}{2}$: likely.

2. $\frac{1}{2}$ is an even chance.

4. Count the objects to find each probability.

Level 1

Baby is a girl

1. About half

Answer: B (even chance)

Level 2

0.29

1. Less than $\frac{1}{2}$

Answer: unlikely

Level 3

$P(\text{yellow})$

1. 6 of 14

Answer: $\frac{3}{7}$

Extension

Green counters

1. 0.25×20

Answer: 5

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LESSON
11.04

Experimental probability

Date _____

Learning intentions

We are learning to:

- compare experimental and theoretical probability

Success criteria

I can:

- ☐ find theoretical probability from equally likely outcomes
- ☐ find experimental (observed) probability from results
- ☐ find expected frequency = probability × number of trials

Key terms Write the definition as your teacher explains it.

Experimental probability

Expected frequency

Trial

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Find the theoretical probability and the expected number.

- a** A coin is tossed 80 times. What is the theoretical P(head)?

- b** What is the expected number of heads in 80 tosses?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Find the theoretical probability.

12 QUESTIONS

REMEMBER Theoretical P = favourable ÷ total (equally likely outcomes).

- 1** P(head) on a coin

- 2** P(4) on a die

- 3** P(A) on spinner ABCD

- 4** P(even) on a die

- 5** P(red) from 3 red, 2 blue

- 6** P(vowel) from A, B, C, D, E

Questions 7–12: find the expected number.

- 7** heads in 50 tosses

- 8** 6s in 30 rolls

- 9** even numbers in 40 rolls

- 10** A on spinner ABCD in 100 spins

- 11** tails in 200 tosses

- 12** a 1 or 2 in 60 rolls

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: find the expected number: probability $\times$ number of trials.

1 an even number on a die in 46 trials

2 a red card in 6 trials

3 a 6 on a die in 108 trials

4 an even number on a die in 8 trials

5 an even number on a die in 44 trials

6 a heart in 72 trials

7 a 1 or 2 on a die in 78 trials

8 heads in 8 trials

9 a 1 or 2 on a die in 27 trials

10 a heart in 40 trials

11 a 6 on a die in 174 trials

12 an even number on a die in 42 trials

13 a heart in 92 trials

14 a red card in 36 trials

15 a red card in 54 trials

16 a heart in 96 trials

17 a 1 or 2 on a die in 45 trials

18 a 6 on a die in 102 trials

19 an even number on a die in 54 trials

20 a 6 on a die in 54 trials

21 a heart in 104 trials

22 a 6 on a die in 72 trials

23 a 1 or 2 on a die in 39 trials

24 an even number on a die in 36 trials

ROUND 2

Questions 25–40: find the experimental probability (relative frequency) as a fraction in simplest form.

25 11 out of 40

26 38 out of 60

27 41 out of 60

28 39 out of 100

29 14 out of 60

30 9 out of 20

31 13 out of 20

32 37 out of 50

33 45 out of 60

34 7 out of 50

35 7 out of 20

36 27 out of 60

37 23 out of 60

38 32 out of 40

39 32 out of 60

40 40 out of 60

YOU DO

On your own: **Easy** Set B. Find the expected frequency.

6 QUESTIONS

1 Heads in 80 tosses

4 A on spinner ABCD in 200 spins

2 6s in 60 rolls

5 Red in 100 draws (3 red, 2 blue, replaced)

3 Even numbers in 50 rolls

6 Tails in 1000 tosses



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Find the observed probability, then compare.

a It landed heads 24 times. What is the observed probability of a head?

Your teacher works this out. Copy every step.

b How does the observed frequency compare with the expected frequency?

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Find the observed (experimental) probability.

6 QUESTIONS

REMEMBER Observed P = number of times it happened ÷ number of trials.

1 24 heads in 80 tosses

3 42 reds in 100 draws

5 130 A's in 400 spins

2 15 sixes in 60 rolls

4 7 wins in 20 games

6 0 fives in 12 rolls

SKILL DRILL 2 Medium basics: keep practising

24 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 24 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1

Questions 1–12: find the experimental probability as a percentage. Compare it with the theoretical probability.

- | | |
|--|--|
| 1 a 6 came up 13 times in 50 rolls
_____ | 2 a 6 came up 17 times in 100 rolls
_____ |
| 3 a 6 came up 22 times in 100 rolls
_____ | 4 a 6 came up 8 times in 20 rolls
_____ |
| 5 a 6 came up 12 times in 40 rolls
_____ | 6 a 6 came up 7 times in 40 rolls
_____ |
| 7 a 6 came up 8 times in 50 rolls
_____ | 8 a 6 came up 21 times in 100 rolls
_____ |
| 9 a 6 came up 18 times in 100 rolls
_____ | 10 a 6 came up 5 times in 50 rolls
_____ |
| 11 a 6 came up 5 times in 40 rolls
_____ | 12 a 6 came up 7 times in 20 rolls
_____ |

ROUND 2 Questions 13–24: find the expected number.

- | | | |
|--|---|--|
| 13 a spinner has 4 equal sectors, 2 shaded; 40 spins land on shaded
_____ | 14 a spinner has 5 equal sectors, 4 shaded; 20 spins land on shaded
_____ | 15 a spinner has 5 equal sectors, 1 shaded; 55 spins land on shaded
_____ |
| 16 a spinner has 8 equal sectors, 3 shaded; 72 spins land on shaded
_____ | 17 a spinner has 8 equal sectors, 5 shaded; 112 spins land on shaded
_____ | 18 a spinner has 5 equal sectors, 1 shaded; 90 spins land on shaded
_____ |
| 19 a spinner has 4 equal sectors, 3 shaded; 76 spins land on shaded
_____ | 20 a spinner has 8 equal sectors, 7 shaded; 120 spins land on shaded
_____ | 21 a spinner has 8 equal sectors, 2 shaded; 64 spins land on shaded
_____ |
| 22 a spinner has 4 equal sectors, 2 shaded; 32 spins land on shaded
_____ | 23 a spinner has 5 equal sectors, 1 shaded; 35 spins land on shaded
_____ | 24 a spinner has 8 equal sectors, 5 shaded; 56 spins land on shaded
_____ |

Level 3 · Challenge

Challenging · 2 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find the expected number, then explain.

- a** A die is rolled 60 times. How many 6s are expected?

Your teacher works this out. Copy every step.

- b** Why do more trials give a better estimate?

Your teacher works this out. Copy every step.

YOU DO

On your own: **Challenging** Set D. Do an experiment. Toss a coin 20 times (or use results from your class).

2 QUESTIONS

REMEMBER Expected frequency = $P \times \text{number of trials}$.

- 1** Record your results in a tally table.

Outcome	Tally	Frequency
Heads		
Tails		

- 2** Find your observed $P(\text{head})$. Compare it with the theoretical probability. Combine with the class results. What do you notice?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What is the theoretical $P(3)$ on a die?

Level 2

How many 3s are expected in 120 rolls?

Level 3

A 3 came up 25 times in 120 rolls. Find the observed $P(3)$.

 Extension · Stretch yourself Worked example

A spinner landed on red 45 times in 150 spins. Estimate how many times it will land on red in 1000 spins.

1. Observed $P(\text{red}) = 45 \div 150 = 0.3$
2. 0.3×1000

Answer: about 300 times

YOU DO

On your own: read the worked example, then try these.

- E1** A drawing pin landed point up 36 times in 60 throws. Estimate how many times it lands point up in 250 throws.

- E2** Is this spinner likely to have equal sectors if there are 4 colours? Explain.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Theoretical $P = \text{favourable} \div \text{total}$ (equally likely outcomes).
 3. Expected frequency = $P \times \text{number of trials}$.

2. Observed $P = \text{number of times it happened} \div \text{number of trials}$.
4. More trials: observed P gets closer to theoretical P .

 Level 1

Theoretical $P(\text{head})$

1. 1 of 2

Answer: $\frac{1}{2}$

 Level 2

Expected heads in 80

1. $\frac{1}{2} \times 80$

Answer: 40

 Level 3

Observed $P(\text{head})$

1. $24 \div 80$

Answer: $\frac{3}{10}$

 Extension

Red in 1000 spins

1. 0.3×1000

Answer: about 300

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
11.05

Complementary events

Date _____

Learning intentions

We are learning to:

- find the probability of complementary events

Success criteria

I can:

- ☐ describe the **complement** of an event (“not” the event)
- ☐ use $P(\text{not } E) = 1 - P(E)$
- ☐ solve problems using complementary events

Key terms Write the definition as your teacher explains it.

Complementary events

Complement

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write the complement and its probability.

- a** Write the complement of selecting a queen from a deck, and its probability.

- b** Write the complement of rolling a multiple of 3 on a die, and its probability.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Write the complementary event.

12 QUESTIONS

REMEMBER The complement of E is “not E”.

- 1** Rolling a 6

- 2** Getting a head

- 3** Drawing a red card

- 4** Winning a game (no draws)

- 5** Choosing a vowel

- 6** Rolling an even number

Questions 7–12: find $P(\text{not } E)$.

- 7** $P(E) = 0.3$

- 8** $P(E) = \frac{1}{5}$

- 9** $P(E) = 45\%$

- 10** $P(E) = 0$

- 11** $P(E) = \frac{3}{8}$

- 12** $P(E) = 0.99$

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: find $P(\text{not } E)$: subtract $P(E)$ from 1.

1 $P(E) = \frac{1}{5}$

2 $P(E) = \frac{1}{4}$

3 $P(E) = \frac{5}{7}$

4 $P(E) = \frac{1}{6}$

5 $P(E) = \frac{3}{4}$

6 $P(E) = \frac{5}{6}$

7 $P(E) = \frac{7}{8}$

8 $P(E) = \frac{1}{12}$

9 $P(E) = \frac{1}{2}$

10 $P(E) = \frac{3}{7}$

11 $P(E) = \frac{1}{20}$

12 $P(E) = \frac{1}{7}$

13 $P(E) = \frac{3}{8}$

14 $P(E) = \frac{14}{15}$

15 $P(E) = \frac{2}{5}$

16 $P(E) = \frac{11}{12}$

17 $P(E) = \frac{2}{15}$

18 $P(E) = \frac{7}{12}$

19 $P(E) = \frac{2}{3}$

20 $P(E) = \frac{5}{8}$

21 $P(E) = \frac{1}{8}$

22 $P(E) = \frac{19}{20}$

23 $P(E) = \frac{5}{12}$

24 $P(E) = \frac{4}{5}$

ROUND 2 Questions 25–40: find $P(\text{not } E)$.

25 $P(E) = 0.54$

26 $P(E) = 0.44$

27 $P(E) = 38\%$

28 $P(E) = 20\%$

29 $P(E) = 56\%$

30 $P(E) = 0.9$

31 $P(E) = 0.02$

32 $P(E) = 45\%$

33 $P(E) = 64\%$

34 $P(E) = 0.99$

35 $P(E) = 43\%$

36 $P(E) = 22\%$

37 $P(E) = 0.51$

38 $P(E) = 0.56$

39 $P(E) = 0.63$

40 $P(E) = 59\%$

YOU DO

On your own: **Easy** Set B. Find $P(\text{not } E)$.

6 QUESTIONS

1 $P(E) = \frac{1}{4}$

4 $P(E) = \frac{5}{12}$

2 $P(E) = 0.3$

5 $P(E) = 0.07$

3 $P(E) = 45\%$

6 $P(E) = 1$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Write the complement and its probability, then find P .

- a** Write the complement of buying the winning ticket out of 550 sold, and its probability.

Your teacher works this out. Copy every step.

- b** Names Ashleigh, Hamish, Sarah, Jake, Hayley: find $P(\text{Hamish})$.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Find each probability.

6 QUESTIONS

REMEMBER E and not E cover all outcomes.

- 1 $P(\text{not a queen})$ from a deck

- 3 $P(\text{not winning})$ with 1 ticket of 550

- 5 $P(\text{not brown})$: 8 brown, 4 black, 3 white

- 2 $P(\text{not a multiple of 3})$ on a die

- 4 $P(\text{not Hamish})$ from the 5 names

- 6 $P(\text{not a heart})$ from a deck

SKILL DRILL 2 Medium basics: keep practising

22 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 22

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–10: describe the complementary event.

1 choosing a vowel

2 winning

3 it rains

4 choosing a red card

5 rolling a 6

6 tossing heads

7 passing the test

8 rolling less than 3

9 rolling an even number

10 choosing a heart

ROUND 2 Questions 11–22: find P(not the colour given) as a fraction.

11 9 red, 6 blue, 5 green: P(not blue)

12 2 red, 9 blue, 8 green: P(not blue)

13 7 red, 5 blue, 1 green: P(not blue)

14 3 red, 9 blue, 5 green: P(not blue)

15 6 red, 6 blue, 2 green: P(not blue)

16 3 red, 5 blue, 3 green: P(not red)

17 1 red, 4 blue, 2 green: P(not blue)

18 3 red, 5 blue, 6 green: P(not red)

19 2 red, 3 blue, 6 green: P(not red)

20 2 red, 9 blue, 5 green: P(not red)

21 2 red, 6 blue, 6 green: P(not blue)

22 3 red, 8 blue, 6 green: P(not green)

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find each probability.

- a** Find $P(\text{a girl's name})$ and $P(\text{not beginning with H})$.

Your teacher works this out. Copy every step.

- b** A bag has 8 brown, 4 black and 3 white marbles. Find $P(\text{not brown})$.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Problems.

4 QUESTIONS

REMEMBER $P(\text{not } E) = 1 - P(E)$.

- 1** The probability it will rain tomorrow is 0.35. What is the probability it will not rain?

- 3** In a class, $P(\text{student has a pet}) = 17/25$. There are 25 students. How many do not have a pet?

- 2** $P(\text{a team wins}) = 0.6$ and $P(\text{draw}) = 0.1$. Find $P(\text{the team loses})$.

- 4** Explain why $P(E) + P(\text{not } E) = 1$.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write the complement of rolling a 2.

Level 2

If $P(E) = 0.8$, find $P(\text{not } E)$.

Level 3

A bag has 5 red, 3 blue and 2 green. Find $P(\text{not red})$.

 Extension · Stretch yourself Worked example

A spinner has $P(\text{red}) = 0.2$, $P(\text{blue}) = 0.45$, $P(\text{green}) = \text{the rest}$. Find $P(\text{not blue})$ and $P(\text{green})$.

1. $P(\text{not blue}) = 1 - 0.45$
2. $P(\text{green}) = 1 - 0.2 - 0.45$

Answer: $P(\text{not blue}) = 0.55$, $P(\text{green}) = 0.35$

YOU DO

On your own: read the worked example, then try these.

- E1 A bag has red, white and blue marbles. $P(\text{red}) = \frac{1}{3}$ and $P(\text{white}) = \frac{1}{4}$. Find $P(\text{blue})$.

- E2 In 200 spins of the spinner above, how many times would you expect NOT green?

Lesson summary

Missed the lesson? Start here.

Remember

1. The complement of E is “not E”.
2. E and not E cover all outcomes.
3. $P(\text{not E}) = 1 - P(E)$.
4. Use it when “not” is easier to count.

 Level 1

Not a queen

1. 4 queens of 52

Answer: $\frac{12}{13}$

 Level 2

Not a multiple of 3

1. $1 - \frac{1}{3}$

Answer: $\frac{2}{3}$

 Level 3

Not brown

1. 7 of 15

Answer: $\frac{7}{15}$

 Extension

Spinner

1. $1 - 0.45$

Answer: 0.55

This space is left blank on purpose. It is the back of your exit ticket.